RESEARCH NOTES

Reading a Spatial Dengue Experiment Carefully

A guide to what a synthetic reaction–diffusion experiment can—and cannot—show about climate-sensitive transmission.

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Article URL: https://skckenneth.github.io/writing/climate-driven-dengue/

The project couples temperature-dependent mosquito dynamics to a spatial reaction–diffusion model. Its experiment matrix is synthetic: it supports comparisons under controlled assumptions, not a forecast for Macau or another specific city.

A small mathematical core

A spatial compartment can be written schematically as

$$\frac{\partial I}{\partial t} = D_I \nabla^2 I + \mathcal{T}(S, I, V, T) - \gamma I,$$

where diffusion represents local movement and $\mathcal{T}$ collects transmission terms that depend on hosts, vectors, and temperature. Long-range dispersal adds a nonlocal process.

How to read the output

The useful question is whether a comparison remains stable when resolution, step size, or uncertain parameters change. The repository therefore includes numerical verification, tests, and sensitivity runs. These establish implementation confidence; empirical calibration is a separate task.

```
# Reproduce the documented synthetic matrix inside the project environment
python run_experiment_matrix.py --tfinal 365 --grid 30 --dt 0.2 --save 30
```

Evidence record: **climate-driven-dengue** · Technical materials are maintained privately.